

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	Y Sai kumar	Reg. No.:	192425218
Section / Batch:	B.Tech AIML	Date:	11-07-2026

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Section 1: Regular Expression Pattern Generation

Student Details:

Name: John Doe

Email: john.doe123@gmail.com

Mobile: +91-9876543210

Password: P@ssw0rd123

Date of Birth: 15/08/2004

Register Number: 23AIML1056

Department: Artificial Intelligence and Machine Learning

For the given student details formulate Regex patterns for the following:

Email ID

Mobile Number

Strong Password

Date of Birth (DD/MM/YYYY)

Register Number

CODE

```
import re
```

```
text = """
```

```
Name: Sai kumar
```

```
Email: saikumar@gmail.com
```

```
Mobile: +91-9123456789
```

```
Password: sai123
```

```
Date of Birth: 22/11/2007
```

```
Register Number: 24AIML5218
```

```
Department: Artificial Intelligence and Data Science
```

```
"""
```

```
print("Email :", re.findall(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", text))
```

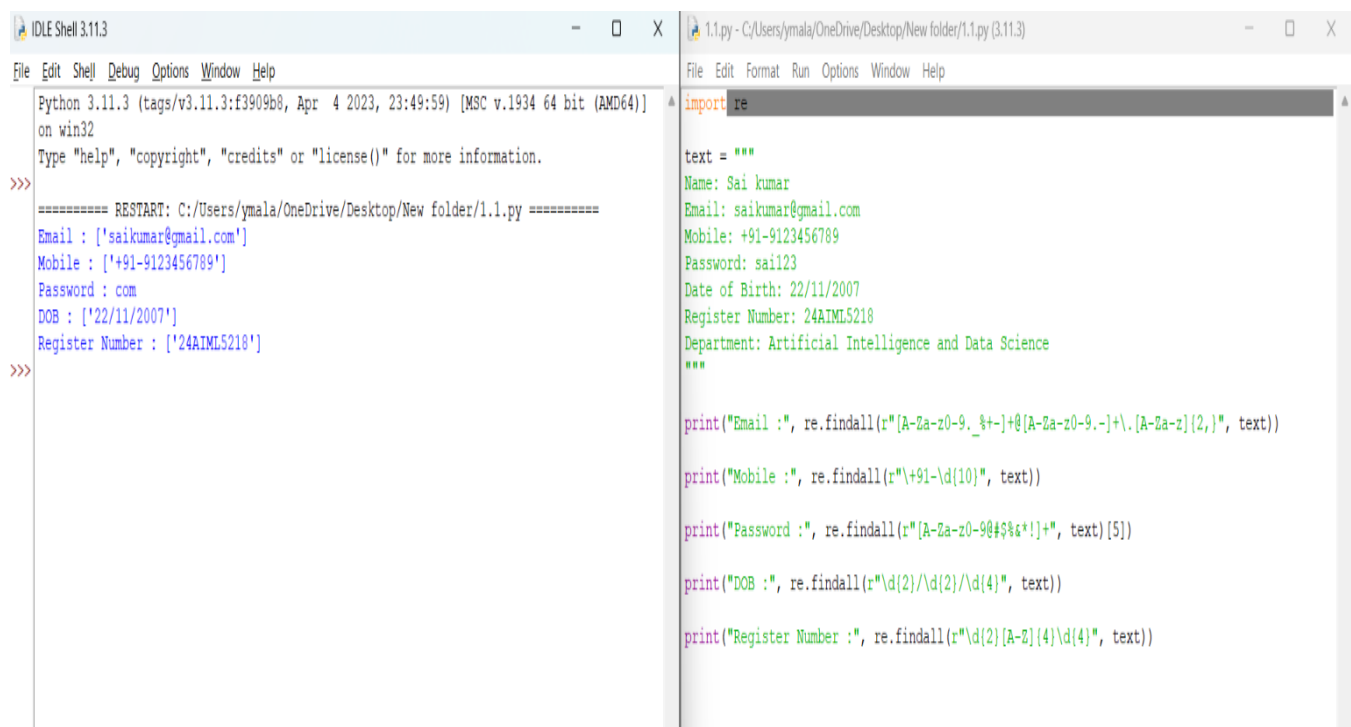
```
print("Mobile :", re.findall(r"\+91-\d{10}", text))
```

```
print("Password :", re.findall(r"[A-Za-z0-9@#%$*!]+", text)[5])
```

```
print("DOB :", re.findall(r"\d{2}/\d{2}/\d{4}", text))
```

```
print("Register Number :", re.findall(r"\d{2}[A-Z]{4}\d{4}", text))
```

```
OUTPUT
```



The screenshot displays two side-by-side windows from an IDE. The left window, titled 'IDLE Shell 3.11.3', shows the execution of a Python script. It starts with a restart message and then prints the results of five regular expression searches on a text string. The output is as follows:

```
>>>
===== RESTART: C:/Users/ymala/OneDrive/Desktop/New folder/1.1.py =====
Email : ['saikumar@gmail.com']
Mobile : ['+91-9123456789']
Password : com
DOB : ['22/11/2007']
Register Number : ['24AIML5218']
>>>
```

The right window, titled '1.1.py - C:/Users/ymala/OneDrive/Desktop/New folder/1.1.py (3.11.3)', shows the source code of the script. It imports the 're' module, defines a text string with student details, and uses five print statements with regular expressions to extract specific information from the text.

```
import re

text = """
Name: Sai kumar
Email: saikumar@gmail.com
Mobile: +91-9123456789
Password: sai123
Date of Birth: 22/11/2007
Register Number: 24AIML5218
Department: Artificial Intelligence and Data Science
"""

print("Email :", re.findall(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", text))

print("Mobile :", re.findall(r"\+91-\d{10}", text))

print("Password :", re.findall(r"[A-Za-z0-9@#%$*!]+", text)[5])

print("DOB :", re.findall(r"\d{2}/\d{2}/\d{4}", text))

print("Register Number :", re.findall(r"\d{2}[A-Z]{4}\d{4}", text))
```

Section 2: Search for a String

Artificial Intelligence (AI) is transforming industries across the world. AI is used in healthcare to assist doctors in diagnosis, in banking to detect fraud, and in education to provide personalized learning experiences. Many companies invest heavily in AI research because AI improves efficiency and enables intelligent decision-making. As AI continues to evolve, professionals with AI skills are in high demand.

Write a Python program using the re module to perform the following operations for the word "AI":

1. Find the first occurrence of the word "AI" using re.search().
2. Display the starting position of the first occurrence.
3. Display the ending position of the first occurrence.
4. Display the total number of times the word "AI" appears in the passage.
5. Print an appropriate message if the word is not found.

CODE

```
import re
```

```
text = """Machine Learning (ML) helps computers learn from data.  
ML is widely used in banking, healthcare, agriculture and robotics.  
Many industries invest in ML because ML increases accuracy and efficiency.  
Students are learning ML to build intelligent applications."""
```

```
word = "ML"
```

```
result = re.search(word, text)
```

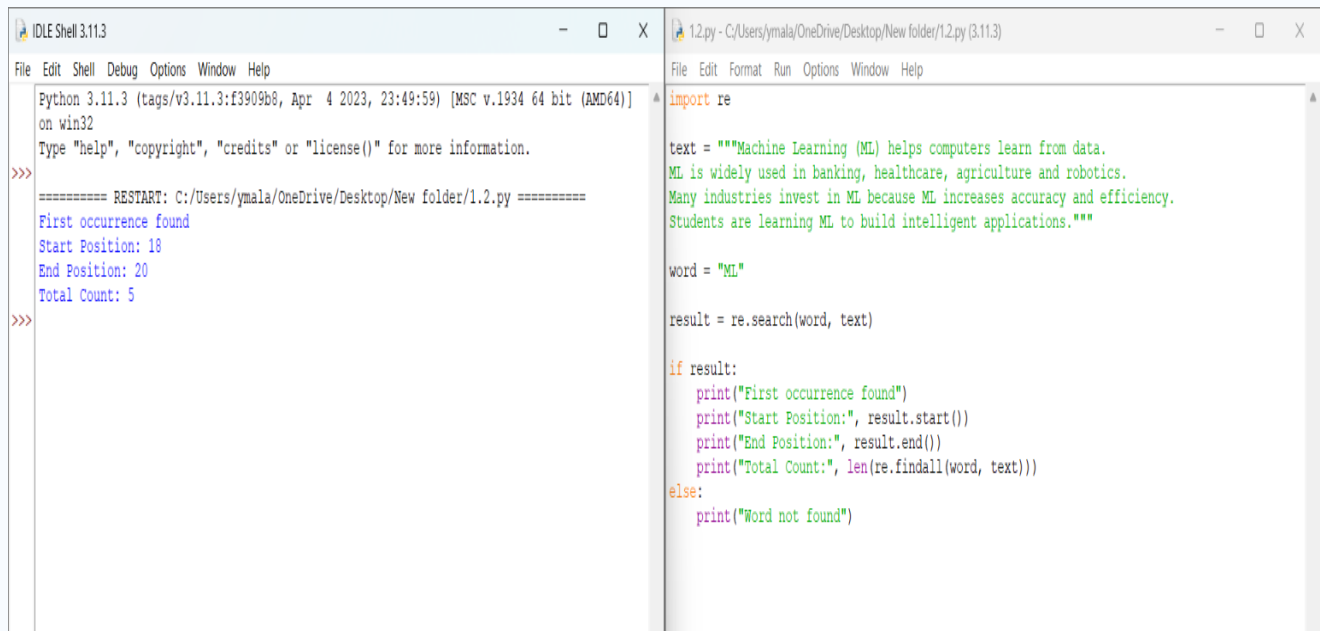
if result:

```
    print("First occurrence found")  
    print("Start Position:", result.start())  
    print("End Position:", result.end())  
    print("Total Count:", len(re.findall(word, text)))
```

else:

```
    print("Word not found")
```

OUTPUT



The screenshot shows a Python IDE with two windows. The left window, titled 'IDLE Shell 3.11.3', displays the output of the program: 'First occurrence found', 'Start Position: 18', 'End Position: 20', and 'Total Count: 5'. The right window, titled '1.2.py - C:/Users/ymala/OneDrive/Desktop/New folder/1.2.py (3.11.3)', shows the Python code used to generate this output. The code imports the re module, defines a text string, sets the word to 'ML', and uses re.search and re.findall to find and count occurrences of the word in the text.

```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)]  
on win32  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
===== RESTART: C:/Users/ymala/OneDrive/Desktop/New folder/1.2.py =====  
First occurrence found  
Start Position: 18  
End Position: 20  
Total Count: 5  
>>>
```

```
import re  
  
text = """Machine Learning (ML) helps computers learn from data.  
ML is widely used in banking, healthcare, agriculture and robotics.  
Many industries invest in ML because ML increases accuracy and efficiency.  
Students are learning ML to build intelligent applications."""  
  
word = "ML"  
  
result = re.search(word, text)  
  
if result:  
    print("First occurrence found")  
    print("Start Position:", result.start())  
    print("End Position:", result.end())  
    print("Total Count:", len(re.findall(word, text)))  
else:  
    print("Word not found")
```

Section 3: Split the documentation into sentences and words

Refer the previous passage and Write a Python program using the re.split() method to perform the following operations:

- Split the passage into sentences using appropriate punctuation (., ?, !) as delimiters.
- Count and display the total number of sentences.
- Split the passage into words by considering one or more whitespace characters as delimiters.
- Count and display the total number of words.
- Display the list of extracted sentences and words.

CODE

```
import re
```

```
text = """Machine Learning (ML) helps computers learn from data.
ML is widely used in banking, healthcare, agriculture and robotics.
Many industries invest in ML because ML increases accuracy and efficiency.
Students are learning ML to build intelligent applications."""
```

```
sentences = re.split(r"[.!?]+", text)
```

```
sentences = [s.strip() for s in sentences if s.strip()]
```

```
words = re.split(r"\s+", text)
```

```
print("Sentences:")
```

```
for s in sentences:
```

```
    print(s)
```

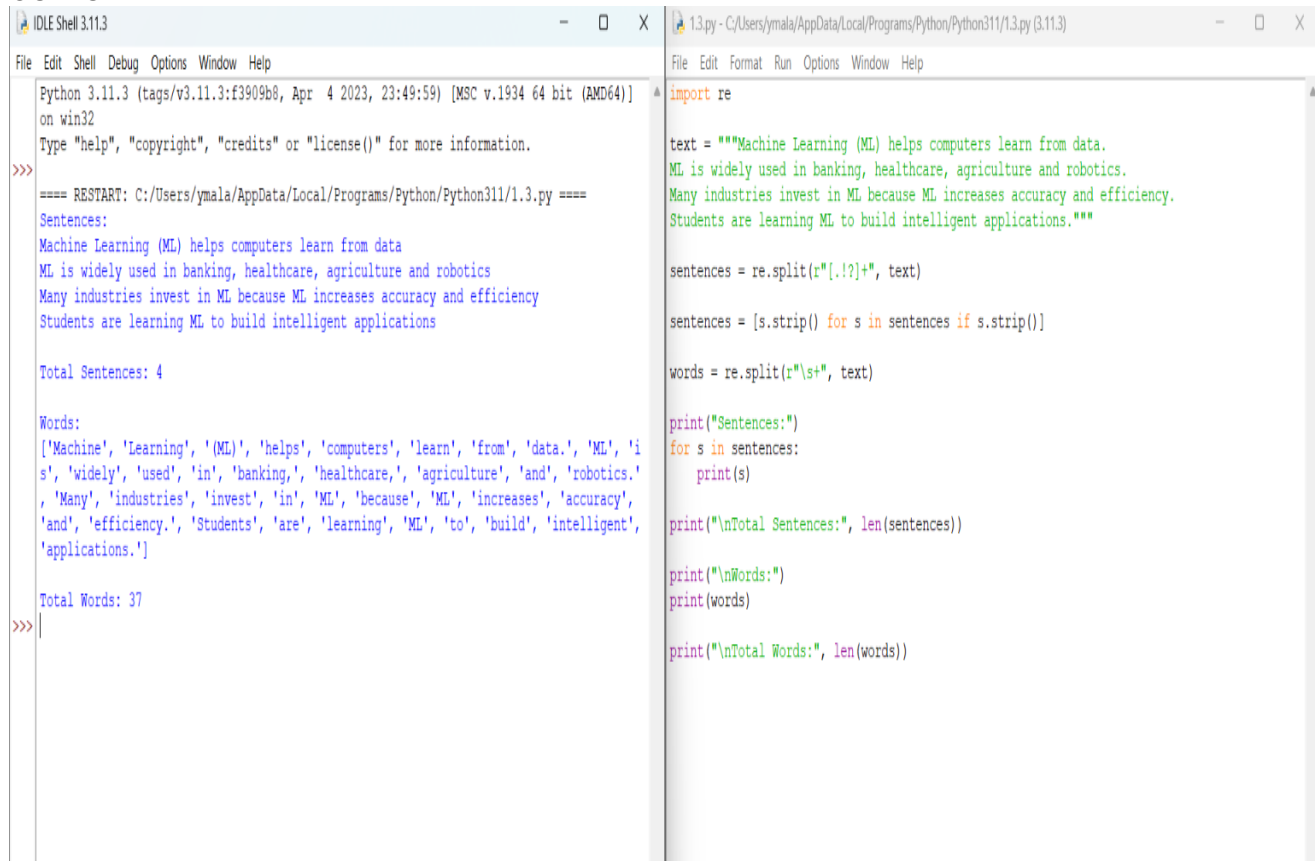
```
print("\nTotal Sentences:", len(sentences))
```

```
print("\nWords:")
```

```
print(words)
```

```
print("\nTotal Words:", len(words))
```

OUTPUT



The screenshot shows two windows of a Python IDE. The left window, titled 'IDLE Shell 3.11.3', displays the output of the script. It shows the text being processed, followed by the sentences split from it, and then the words split from the sentences. The output indicates 4 sentences and 37 words. The right window, titled '1.3.py - C:/Users/ymala/AppData/Local/Programs/Python/Python311/1.3.py (3.11.3)', shows the source code of the script. The code imports the 're' module, defines a text string, splits it into sentences using a regular expression, strips whitespace from the sentences, splits the sentences into words, and prints the results.

```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)]
on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
==== RESTART: C:/Users/ymala/AppData/Local/Programs/Python/Python311/1.3.py ====
Sentences:
Machine Learning (ML) helps computers learn from data
ML is widely used in banking, healthcare, agriculture and robotics
Many industries invest in ML because ML increases accuracy and efficiency
Students are learning ML to build intelligent applications

Total Sentences: 4

Words:
['Machine', 'Learning', '(ML)', 'helps', 'computers', 'learn', 'from', 'data.', 'ML', 'i
s', 'widely', 'used', 'in', 'banking,', 'healthcare,', 'agriculture', 'and', 'robotics.'
, 'Many', 'industries', 'invest', 'in', 'ML', 'because', 'ML', 'increases', 'accuracy',
'and', 'efficiency.', 'Students', 'are', 'learning', 'ML', 'to', 'build', 'intelligent',
'applications.'].
Total Words: 37
>>>
```

```
1.3.py - C:/Users/ymala/AppData/Local/Programs/Python/Python311/1.3.py (3.11.3)
File Edit Format Run Options Window Help
import re

text = """Machine Learning (ML) helps computers learn from data.
ML is widely used in banking, healthcare, agriculture and robotics.
Many industries invest in ML because ML increases accuracy and efficiency.
Students are learning ML to build intelligent applications."""

sentences = re.split(r"[.!?]+", text)

sentences = [s.strip() for s in sentences if s.strip()]

words = re.split(r"\s+", text)

print("Sentences:")
for s in sentences:
    print(s)

print("\nTotal Sentences:", len(sentences))

print("\nWords:")
print(words)

print("\nTotal Words:", len(words))
```

Section 4: Email and Strong Password Validation

A website requires users to register by providing an Email ID and a Strong Password. Write a Python program using the re module to validate the following inputs.

Validation Criteria:

Email ID

- Must begin with one or more letters or digits.
- May contain ., _, %, +, or - before the @ symbol.
- Must contain a valid domain name.
- Must end with a valid domain extension such as .com, .org, .edu, or .in.

Strong Password

- Must contain at least 8 characters.
- Must contain at least one uppercase letter.
- Must contain at least one lowercase letter.

- Must contain at least one digit.
- Must contain at least one special character from @, #, \$, %, &, !, or *.
- Must not contain any whitespace characters.

CODE

```
import re
```

```
email = "student2026@outlook.com"
password = "Study@2026"
```

```
email_pattern = r"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$"
```

```
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#$$%&!*])[A-Za-z\d@#$$%&!*]{8,}$"
```

```
if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
```

```
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

OUTPUT

The screenshot shows two windows from the Python IDLE 3.11.3 application. The left window, titled 'IDLE Shell 3.11.3', displays the output of the program: 'Valid Email' and 'Strong Password'. The right window, titled '1.3.py - C:/Users/ymala/OneDrive/Desktop/New folder/1.3.py (3.11.3)', shows the source code that was executed. The code defines email and password variables, creates regular expression patterns for email and password validation, and uses if statements with re.fullmatch() to check the variables against these patterns, printing the appropriate result.

Section 5: Compile Once, Validate Many

A company receives 1,000 email IDs from its customers and needs to validate each one efficiently. Instead of compiling the same regular expression for every email, write a Python program using the re.compile() method to compile the pattern once and reuse it to validate all the email IDs.

CODE

```
import re
```

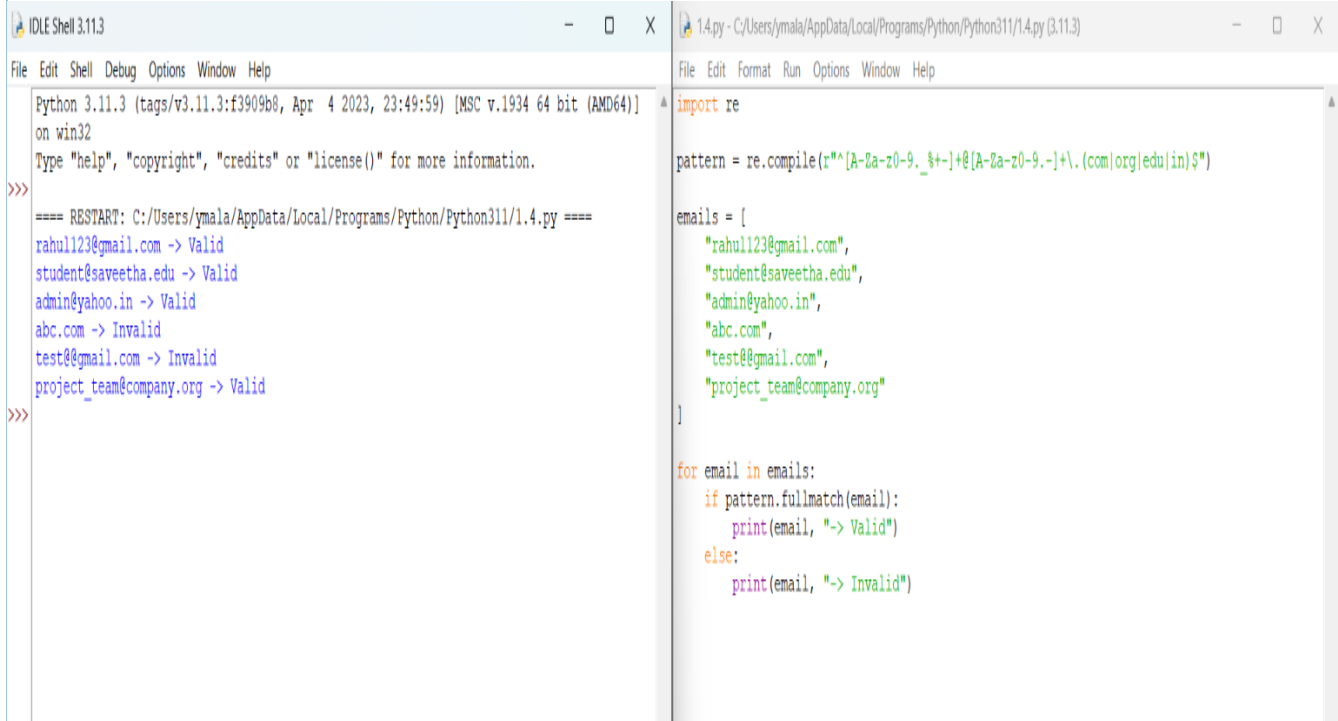
```
pattern = re.compile(r"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$")
```

```
emails = [
    "rahul123@gmail.com",
    "student@saveetha.edu",
    "admin@yahoo.in",
```

```
"abc.com",  
"test@@gmail.com",  
"project_team@company.org"  
]
```

```
for email in emails:  
    if pattern.fullmatch(email):  
        print(email, "-> Valid")  
    else:  
        print(email, "-> Invalid")
```

OUTPUT



The image shows two side-by-side windows from a Python IDE. The left window, titled 'IDLE Shell 3.11.3', displays the output of a Python script. It shows a restart message and a list of email addresses being tested against a pattern. The results are: 'rahul123@gmail.com' is Valid, 'student@saveetha.edu' is Valid, 'admin@yahoo.in' is Valid, 'abc.com' is Invalid, 'test@@gmail.com' is Invalid, and 'project_team@company.org' is Valid. The right window, titled '1.4.py - C:/Users/ymala/AppData/Local/Programs/Python/Python311/1.4.py (3.11.3)', shows the source code of the script. It imports the 're' module, defines a regular expression pattern for email validation, creates a list of email addresses, and uses a loop to check each email against the pattern, printing the result.

```
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)]  
on win32  
Type "help", "copyright", "credits" or "license()" for more information.  
>>>  
==== RESTART: C:/Users/ymala/AppData/Local/Programs/Python/Python311/1.4.py ====  
rahul123@gmail.com -> Valid  
student@saveetha.edu -> Valid  
admin@yahoo.in -> Valid  
abc.com -> Invalid  
test@@gmail.com -> Invalid  
project_team@company.org -> Valid  
>>>
```

```
import re  
  
pattern = re.compile(r"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$")  
  
emails = [  
    "rahul123@gmail.com",  
    "student@saveetha.edu",  
    "admin@yahoo.in",  
    "abc.com",  
    "test@@gmail.com",  
    "project_team@company.org"  
]  
  
for email in emails:  
    if pattern.fullmatch(email):  
        print(email, "-> Valid")  
    else:  
        print(email, "-> Invalid")
```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____